Do redundancy-driven inferences arise automatically during comprehension?

Growing evidence suggests that expectations during comprehension are shaped not only by what is contextually plausible but also by what a cooperative speaker may choose to mention explicitly. However, what happens when such informativity-based expectations are violated remains underexplored. Prior work showed that informational redundancy can be "repaired" by drawing inferences about event typicality. For example, "John went grocery shopping. He paid the cashier!" can elicit the inference that it is atypical for John to pay [1]. Beyond this, an intriguing yet largely unexplored possibility is that comprehenders may also infer the listener's knowledge state (e.g., the listener is a novice to whom even ordinary events seem noteworthy). Furthermore, the cognitive mechanisms underlying such inferences remain unclear: while Relevance Theory [2] predicts that they are automatic, Default Theory [3] predicts that they are non-automatic.

Exp 1 (sentence completion, 48 L1 English speakers from the UK) used a background story featuring three characters—two UK natives (Grace and Alice) and one newcomer to the UK from China (Xinyi; Table 1). In each item, Grace/Alice produced either a redundant or a non-redundant utterance (Table 2, 12 targets, 36 fillers, in English). The non-redundant utterances were informative for both listeners, while the redundant ones involved common knowledge about UK life, and were thus redundant for Grace/Alice but potentially novel for Xinyi. Each utterance was followed by "Grace/Alice said" and a blank for participants to complete. If redundancy-driven inferences are automatic, participants may infer that the redundant (but not the non-redundant) utterances are directed at Xinyi and indicate this in their completions. **Results**: two raters judged whether such an indication was included in each completion ($\kappa = 0.96$). Indications that the utterance was directed at Xinyi were more frequent in the redundant condition (32.64%, n = 94) than in the non-redundant condition (0.35%, n = 1), $p = .011$ (using GLMM). By contrast, participants seldom indicated a UK addressee in either condition (n = 2).

Exp 2 (word recognition, 48 L1 English speakers from the UK) used a similar background story and stimuli as Exp 1 (with different character names; Tables 1–2, 12 targets, 24 fillers, in English). Each item was followed by three words, and participants judged whether each word had appeared in the text. For critical items, these were the unmentioned Chinese and UK characters, and a non-name (e.g., Latte). Again, if redundancy-driven inferences are automatic, participants may infer that redundant utterances are directed at the Chinese character, creating the illusion that her name appeared, leading to higher false recognition rates compared to the UK character. Even when the Chinese character is not falsely recognised, interference from this illusion may result in longer RTs. **Results**: for both false memory rates and RTs of correct recognitions, there was an interaction between redundancy and character type, ps < .001 (using LMM, Fig 1), due to higher false memory rates and longer RTs of correct recognitions for the Chinese character than the UK characters, in the redundant (ps < .001) but not the non-redundant condition (ps > .18).

The current results provide novel evidence that redundancy-driven inferences are automatic during comprehension. These findings support Relevance Theory [2] and extend previous work on informativity-based expectations by showing that listeners do not simply register a redundant utterance, but also use redundancy as a cue to infer communicative intent and shared knowledge.

Table 1. Exp 1 & Exp 2 background story

It is the beginning of a new academic year at the University [blind for review]. The campus is filled with students from all over the world. Grace, Alice, and Xinyi are three first-year undergraduates in the School of Psychology. Grace and Alice were born and raised in the UK, while Xinyi was born and raised in China and has only been in the UK for a week. They met each other during the welcoming week, and found that they have a lot in common. They soon became friends and now spend a lot of time together every day.

Note. In Exp 2, the character names were changed to Chloe, Ellie, and Grace. The three names were fully counterbalanced across six versions of the background story and stimuli.

Table 2. Exp 1 & Exp 2 exemplar stimuli

Exp 1	Alice, Xinyi, and Grace went to Coates Café for lunch, where they had a variety of foods to choose from, including chicken gyros, jacket potatoes, and baked beans. "Jacket potatoes have been a popular dish for many years,"$_{\text{redundant}}$/"I had the chicken gyros last time and they were really good,"$_{\text{non-redundant}}$ Alice said __.
Exp 2	The three friends went to Coates Café for food, and there was a variety of choices, including chicken gyros, jacket potatoes, and baked beans. "Jacket potatoes have been a popular dish for many years,"$_{\text{redundant}}$/"I had the chicken gyros and they were really good,"$_{\text{non-redundant}}$ Grace said. Recognition words: Chloe, Ellie, Latte

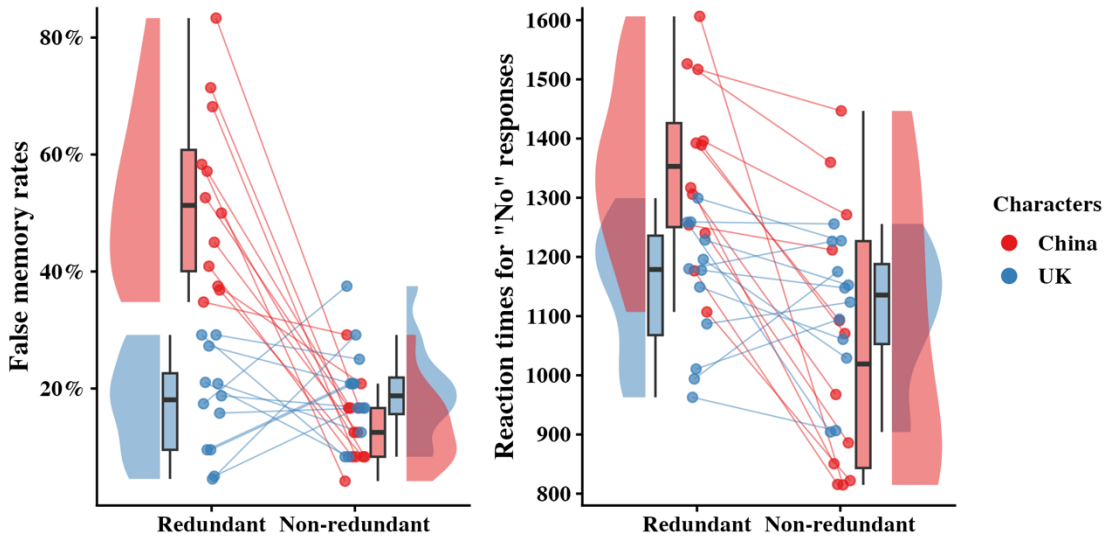

Fig 1. Exp 2 false memory rates and RTs for correct recognitions by item and by condition

References

[1] Kravtchenko, E. & Demberg, V. (2022). Informationally redundant utterances elicit pragmatic inferences. *Cognition*, *225*, Article 105159. [2] Sperber, D., & Wilson, D. (1986). *Relevance: Communication and Cognition*. Harvard University Press. [3] Levinson, S. C. (2000). *Presumptive Meanings: The Theory of Generalized Conversational Implicature*. MIT Press.